4.5.2.2 RQ-4 Global Hawk (1998)

The RQ-4 Global Hawk, developed by Northrop Grumman for the United States Air Force (USAF), represents one of the most advanced and enduring high-altitude long-endurance (HALE) unmanned aerial vehicles (UAVs) of the third generation. Introduced in 1998, the Global Hawk was designed to provide strategic, near-real-time intelligence, surveillance, and reconnaissance (ISR) capabilities over vast areas of interest, operating at altitudes and durations beyond the reach of conventional manned aircraft. It became a cornerstone of modern aerial intelligence operations, combining autonomous navigation, multi-sensor integration, and satellite-based communication into a single, globally deployable platform (Kompała, 2014).

Design and technical overview

The RQ-4 Global Hawk features a composite high-aspect-ratio airframe optimized for aerodynamic efficiency and endurance. With a wingspan of approximately 40 meters (131 feet) and a length of 14.5 meters (47.6 feet), it can operate continuously for over 36 hours at altitudes exceeding 60,000 feet, effectively placing it above commercial air traffic and most weather systems. Powered by a single Rolls-Royce AE 3007H turbofan engine, the UAV achieves a cruise speed of around 570 km/h (355 mph), combining fuel efficiency with persistent operational capability (Hassell Jr, 2009).

The Global Hawk's sensor suite is among the most advanced of its generation. It includes a synthetic aperture radar (SAR) with ground moving target indicator (GMTI) capabilities, electro-optical (EO) and infrared (IR) cameras, and signals intelligence (SIGINT) systems. This combination enables day-and-night, all-weather reconnaissance, allowing the aircraft to image and track ground targets across hundreds of kilometers in a single pass. These payloads are integrated into a modular sensor bay,